Memento

Season: 3 · Behavioral · Category: GoF · Level: Intermediate

Intent

- Capture and restore an object's internal state.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Video game save slot.

When to Use

- See Intent.
- The design force named in Intent shows up more than once.

When Not to Use

- Overkill for one-off behavior.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.gof.behavioral.memento;

import java.util.ArrayDeque;
import java.util.Deque;

public final class TextEditor {
    private String text = "";

    public void type(String chunk) { text += chunk; }
    public String text() { return text; }

    public Memento save() { return new Memento(text); }
    public void restore(Memento m) { text = m.state(); }

    public record Memento(String state) {}

    /** Caretaker */
    public static final class History {
        private final Deque<Memento> stack = new ArrayDeque<>();
        public void push(Memento m) { stack.push(m); }
        public Memento pop() { return stack.pop(); }
        public boolean isEmpty() { return stack.isEmpty(); }
    }
}
```

Modern Java 25 / Functional

- Java 25 pattern matching often replaces Visitor/Interpreter boilerplate.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/behavioral/memento/
```

Pitfalls

- Prefer sealed + switch over classic Visitor in Java 25.
- Strategy as a function value.
- Observer: unsubscribe to avoid leaks.

Interview Q&A

- Q: Key interview angle for Memento? A: Name the force (undo, open/closed, open for extension) and a modern alternative.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Capture and restore an object's internal state.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Video game save slot.
  [0:40] Intent — State the design force in one sentence: Capture and restore an...